Tutorial: How to create Web Application using Python and GPTs
Continuous Assessment 2 (CA-2): SOP
Course: Machine Learning & ANN (CS-303B)

Overview

- **Assessment Type:** CS-303B Project Exhibition
- **Total Marks:** 6 Marks
- **Objective:** To enable students to translate theoretical knowledge into practical AIML web applications.

Syllabus Coverage

The project exhibition will cover topics from the following units as per the course curriculum:

- **Unit 1:** Understanding the Basics of ML (Data Pre-processing, KNN, Regression, SVM, and Performance Metrics)
- **Unit 2:** Feature Engineering, Outlier Detection (Isolation Forest), and Unsupervised Learning (Clustering, PCA)
- **Unit 3:** Ensemble Learning (Random Forest, Stacking), Hyperparameter Tuning, and AutoML Frameworks

Implementation Guidelines

1. **Project Assignment:** Students will be assigned their specific project topics 15 days prior to the scheduled exhibition date.
2. **Team Formation:** The assessment will be conducted in teams consisting of **2 students** each.
3. **Evaluation Process:** The exhibition will feature **cross-department judging** to ensure an objective and diverse evaluation of the implemented AIML solutions.

Project Resources

Students are encouraged to utilize the following resources for dataset acquisition and reference:

- **Dataset Repository:** UCI Machine Learning Repository
- **Indian Open Datasets**: India Open Dataset,AIkosh.
- **Tools:** Vscode, dataset, github account, streamlit account.

Tutorial

Step-1 : Open Gemini Google Gemini.

Step-2 : Enter the following prompt into gemini: create a ML pipeline using streamlit dashboard that does the following in sequence The application to show horizontal expansion of the steps with more graphically appealing and make use of the best esthetics for dashboard 1 Initially the application will ask for the type of problem to be solved ClassificationRegression 2 Input Data Input data from UI and ask user to select the target feature Show the overall data shape using PCA also allow user to select features and see the shape of the data You can also use plotly for this 2 Exploratory Data Analysis EDA Perform EDA and provide the results on the UI 3 Data Engineering and Cleaning Provide options to do data engineering with methods like mean median mode and compute outliers using IQR Isolation Forest DBSCAN and OPTICS If there are outliers then ask user if he wants to remove them Give option to user so that he can delete these data from UI 4 Feature Selection Perform feature selection step using methods like variance threshold correlation information gain only with target 5 Data Split Split the data into training and testing samples 6 Model Selection ask user to select the model from the list linear regression SVM with kernel options kmeans random forest 7 Model Training and KFold Validation Ask for the value of K before this step 8 Performance Metrics with checking for overfitting and underfitting 9 At the end give option to tune hyper parameters for the mentioned models and its effect on the model performance using gridsearch or randomsearch

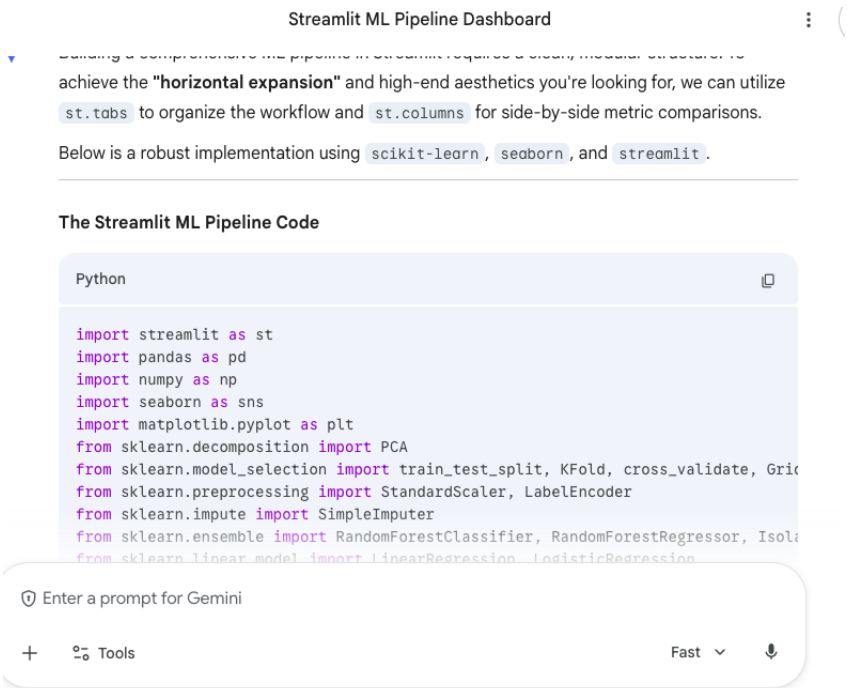

Figure 1: default

Step-3 : Copy and paste the resultant code in IDE and install any dependecies asked for. Save the code with the name pipeline.py Note: only use offline IDE's like vscode

for this.

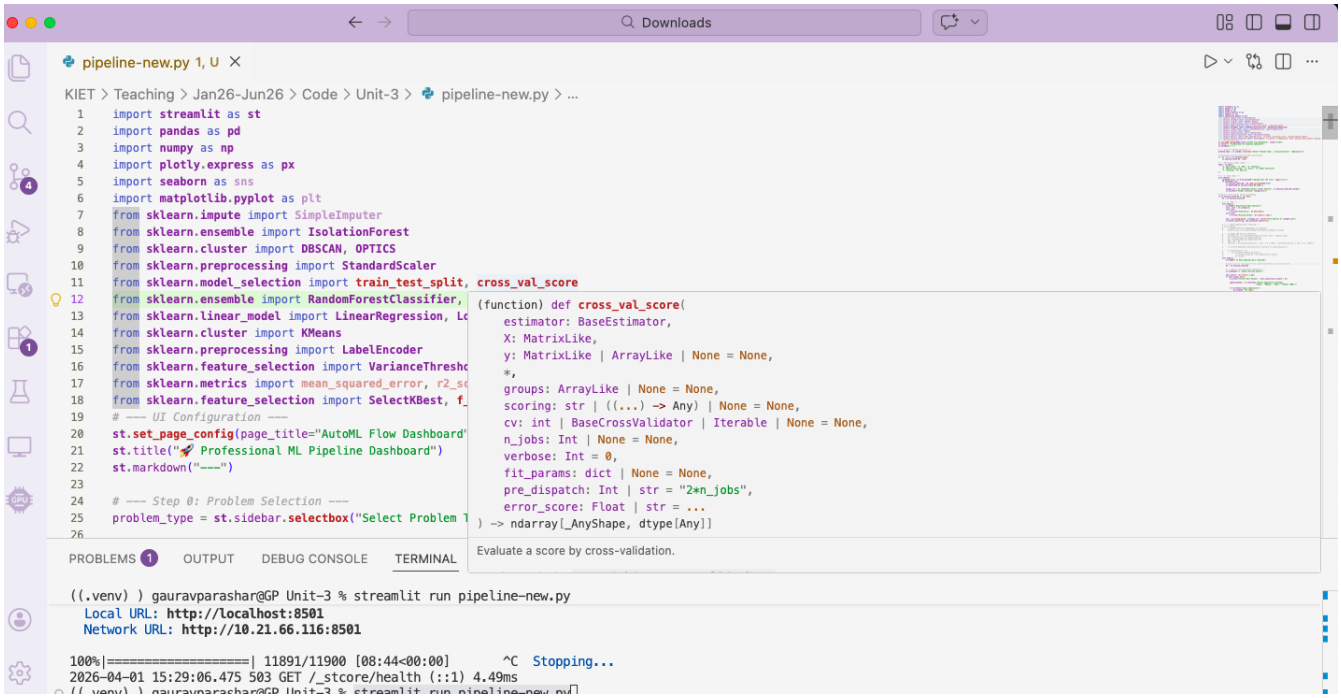

Figure 2: VS Code View

Step-4 : To execute the code run the following command:

```
streamlit run pipeline.py
```

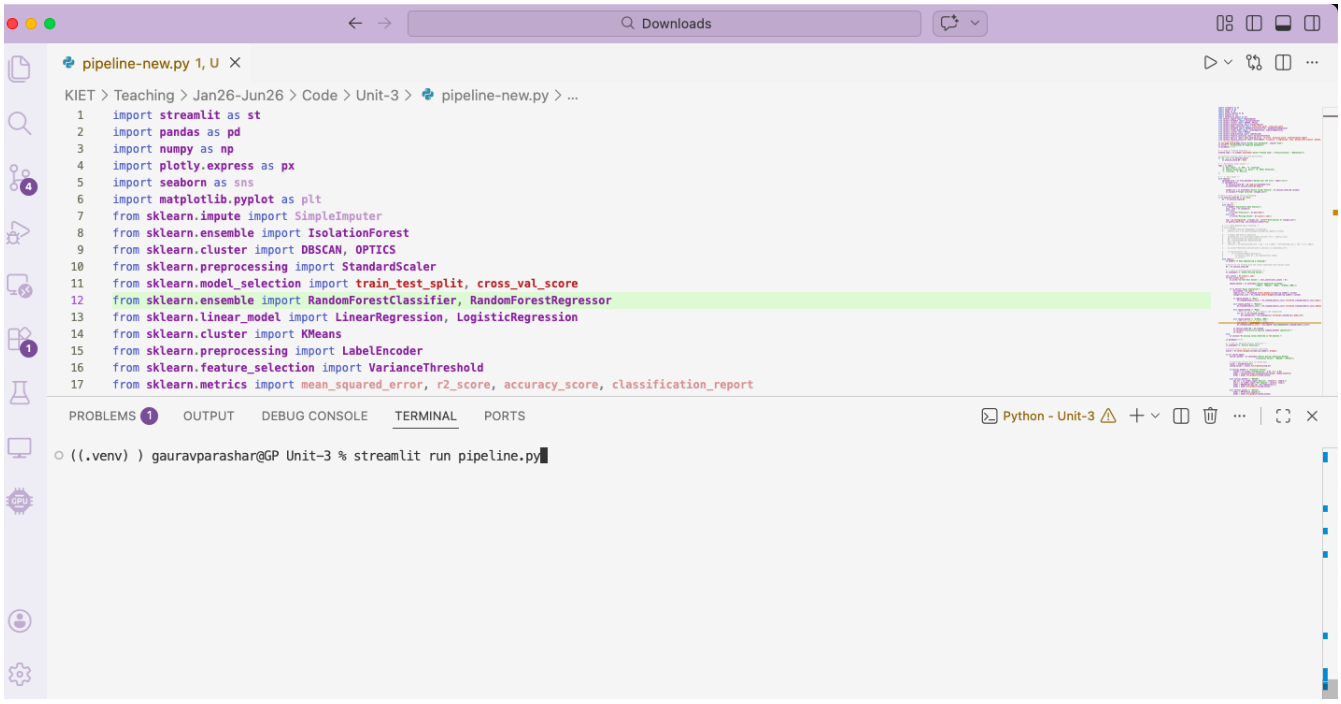

Figure 3: VS Code Terminal View

Step-5 : After the command executes successfully, then the application will automatically open's the webpage in the default browser.

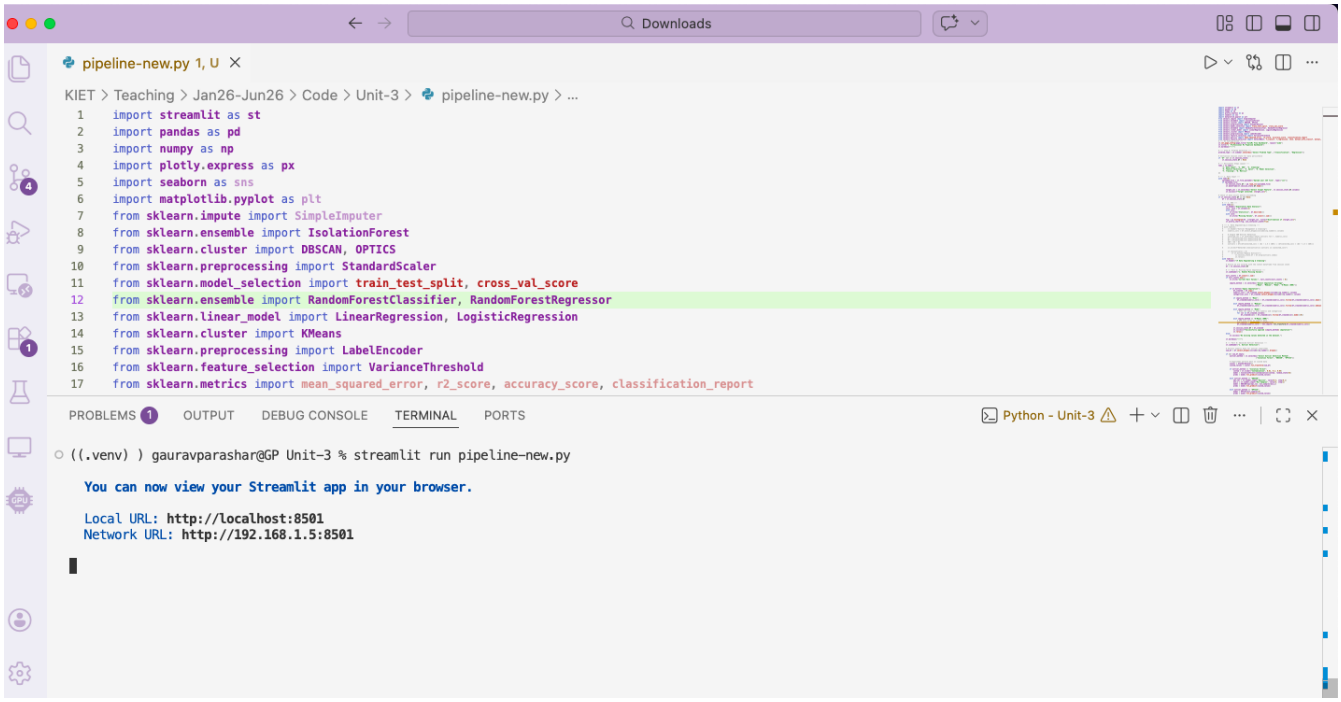

Figure 4: VS Code Terminal View

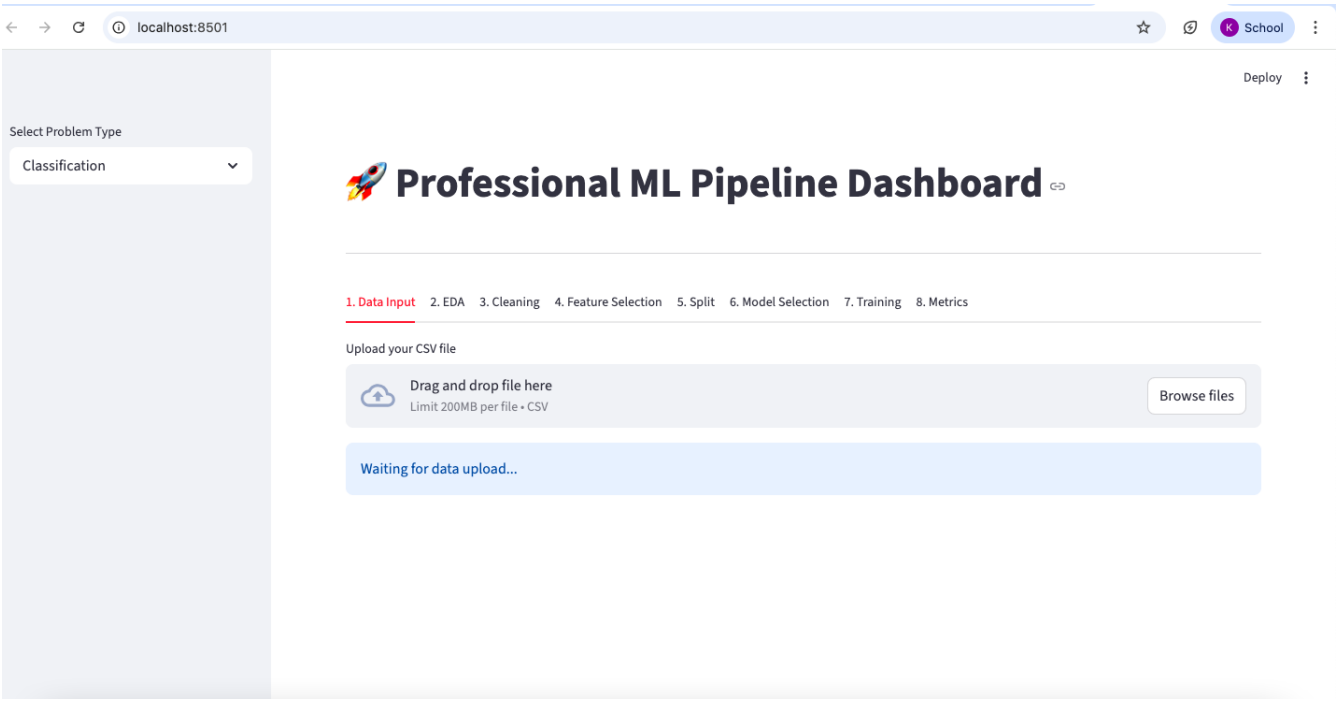

Figure 5: Application running inside browser view

Step-5 : If everything worked fine and you are satisfied with the code working and all the features are correctly performing the task. Then open streamlit website and create an account by clicking on the top right corner icon with text "Free".

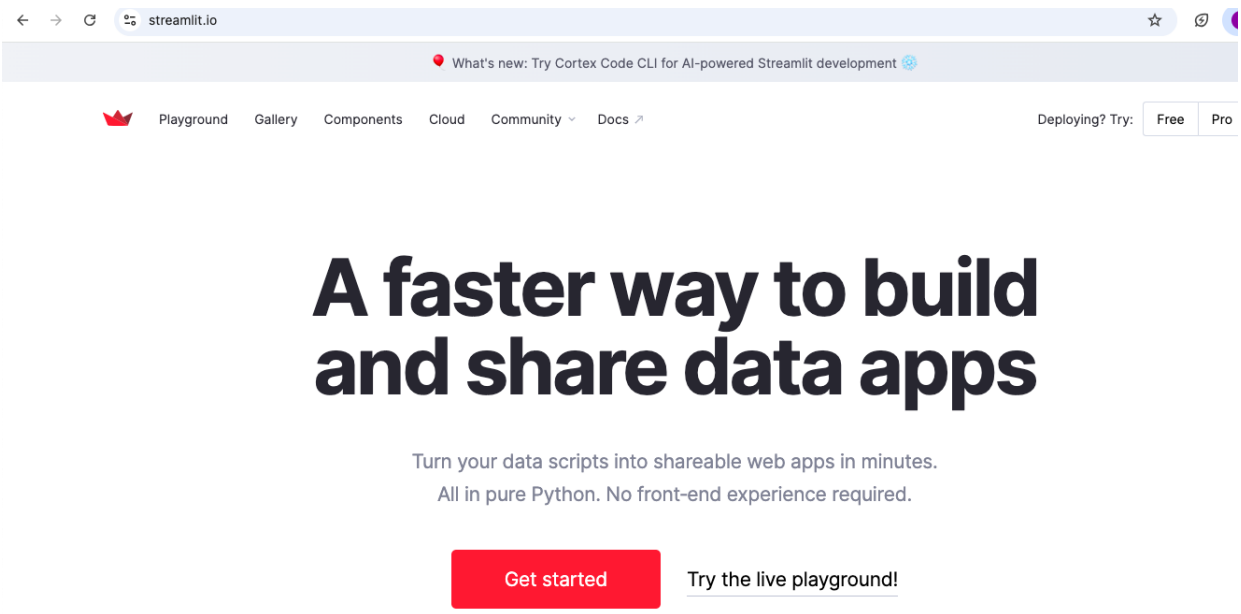

Figure 6: Streamlit Website

Step-6 : Create an account either using Google of Github.

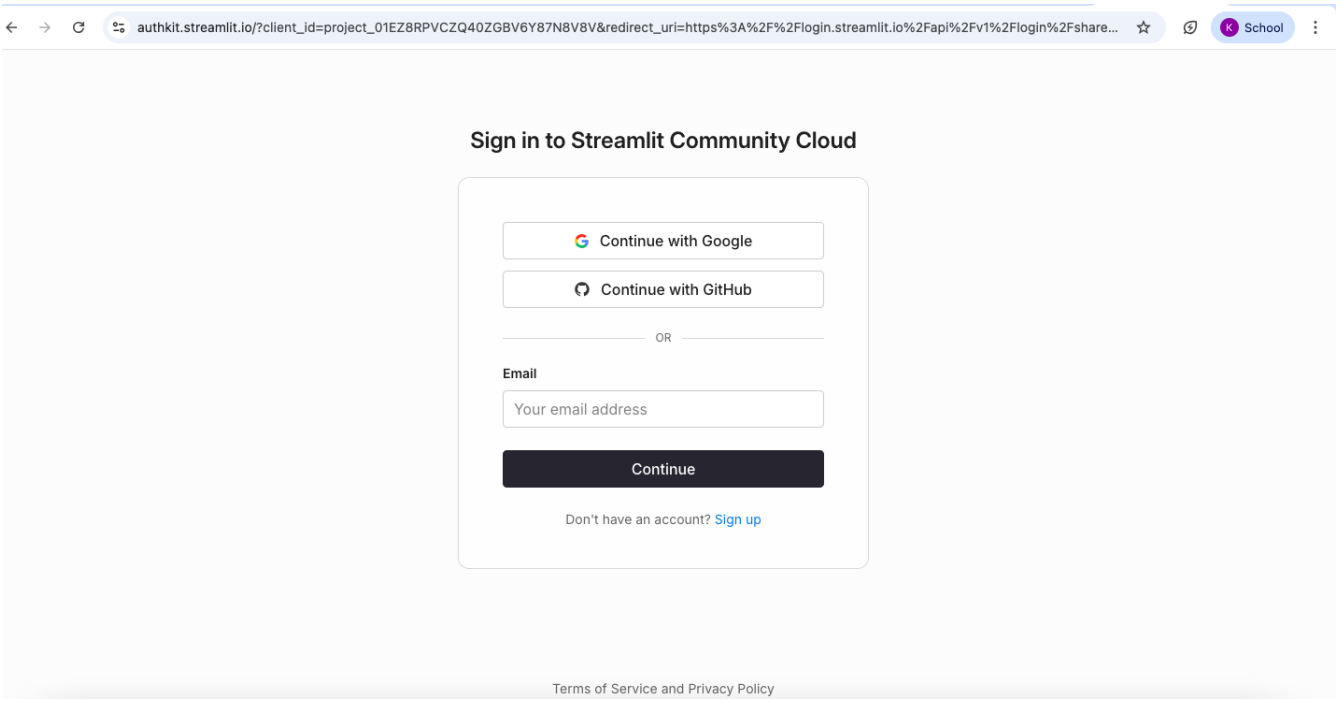

Figure 7: Streamlit Website: Account creation

Step-7 : After successful account creation.

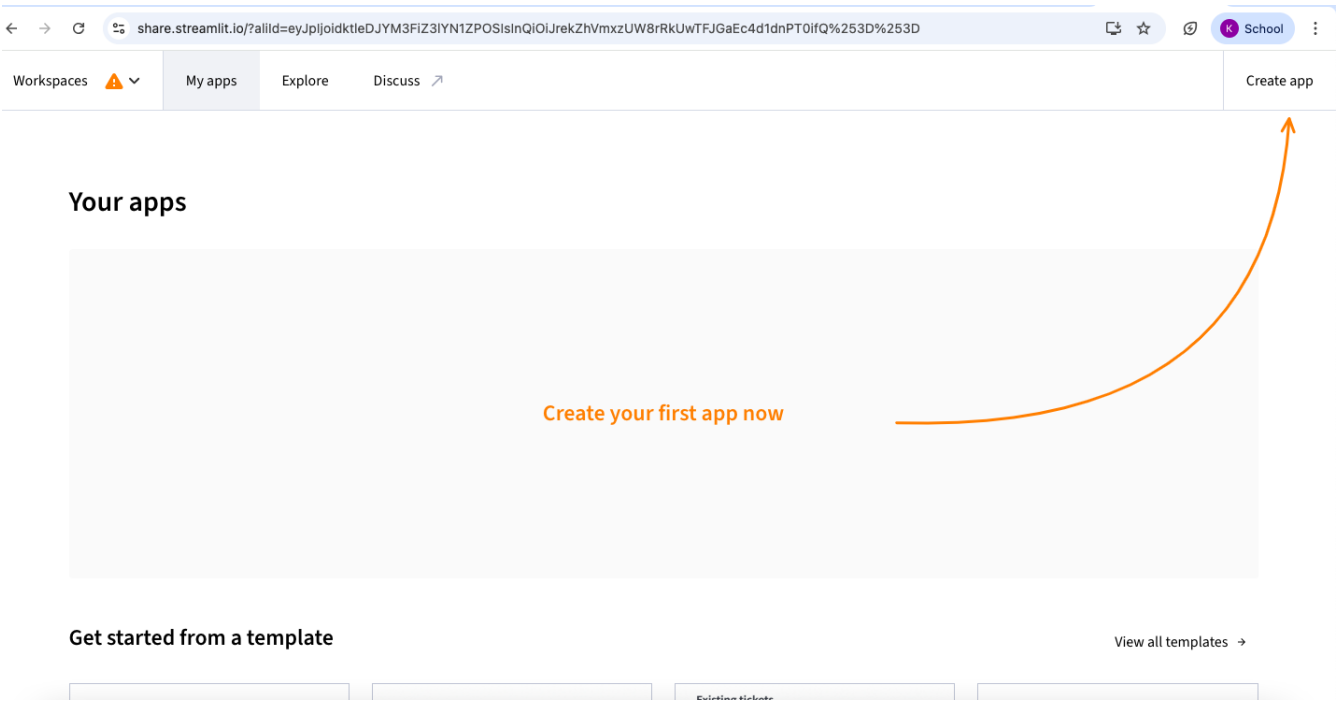

Figure 8: Streamlit Website: App Creation UI

Step-8 : After successful account creation. Click on "Create app" icon on top right corner of the screen.

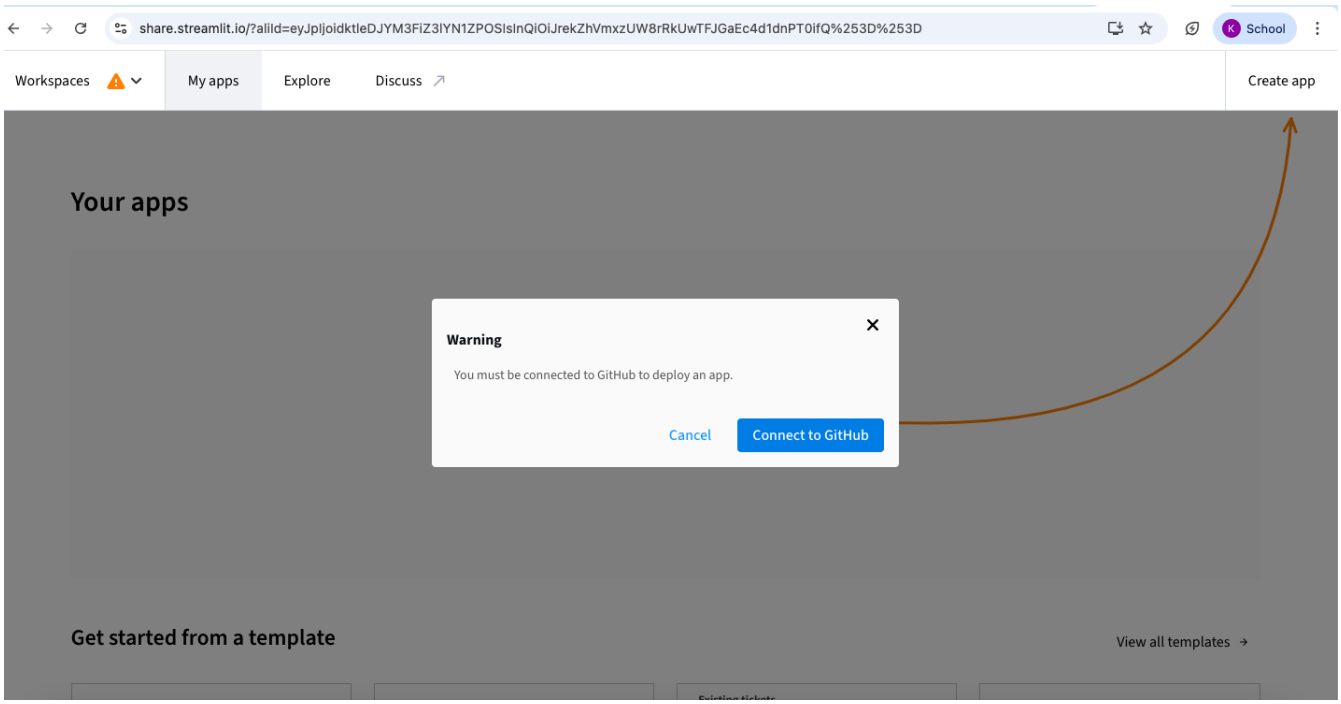

Figure 9: Streamlit Website: App Creation UI

Step-9 : Connect to your Github Account.

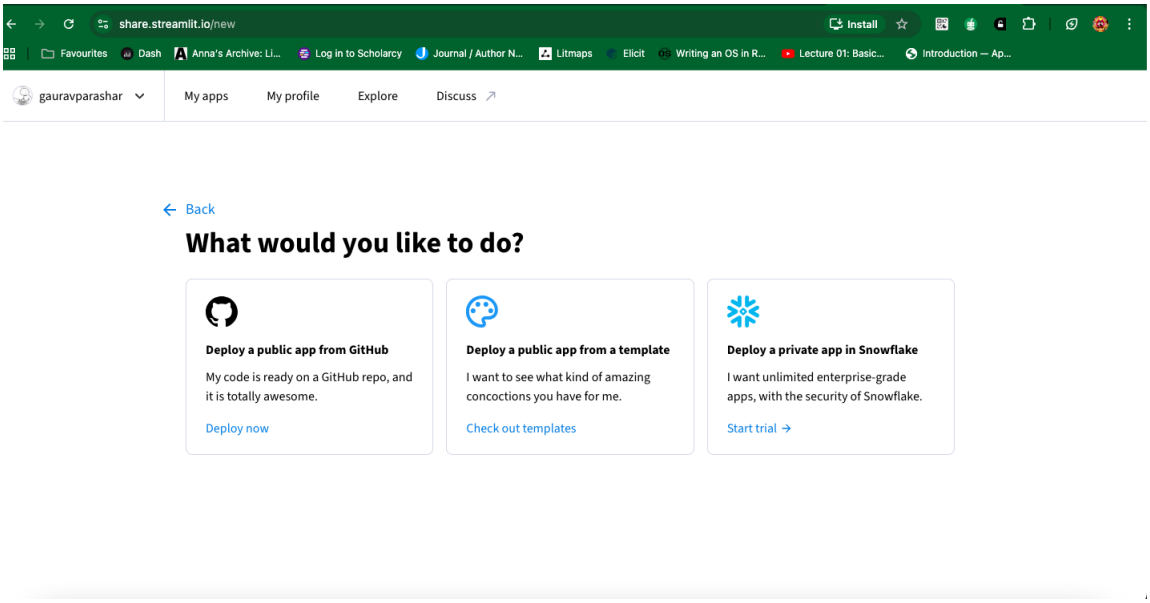

Figure 10: Streamlit Website: App Creation UI

Step-10 : Upload your application on github and make it public.

Step-11 : Enter the following details from your github repository in the fields as mentioned and then click deply. Deployment will give you a url that you can click and test your application. For more information on deployment check the documentation.

After you have understood the working then play with the code and make it more robust, so that it can be used as a full web application.

Happy Coding!!